

Lecture 2: Graphical Analysis, Local and Global Optima

MA3236 Non-Linear Programming

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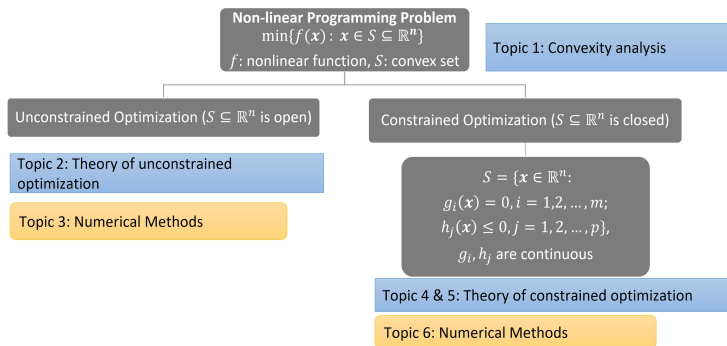
Quick review

- Optimization is not new and has been ever growing.
- Optimization problem is prevalent; it describes mathematically what you want, what you have, and what you want to change.

$$\begin{array}{ll}\min_{\mathbf{x}} & f(\mathbf{x}) \\ \text{s. t.} & \mathbf{x} \in S \subseteq \mathbf{R}^n\end{array}$$

- Unconstrained ($S = \mathbf{R}^n$) vs constrained (general $S \subseteq \mathbf{R}^n$); open/closed is a separate property.
- Minimize vs maximize (symmetry).
- Unboundedness.

Topics



In this lecture, we will learn

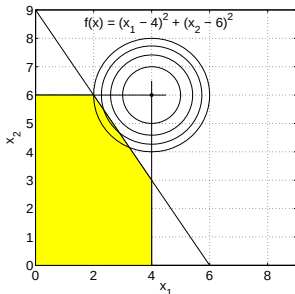
- graphical methods to solve simple optimization models (in $\mathbf{R}^2$);
- notion of local vs global minimizers;
- conditions that guarantee the existence of global optimizers.

Example 2.1 (An NLP (in $\mathbf{R}^2$) with linear constraints but nonlinear objective function.)

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) := (x_1 - 4)^2 + (x_2 - 6)^2 \\ \text{subject to} & x_1 \leq 4 \\ & x_2 \leq 6 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0.\end{array}$$

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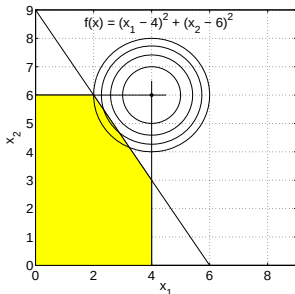


- $(x_1 - 4)^2 + (x_2 - 6)^2 = r^2$ ($r > 0$) describes a circle with center $(4, 6)^T$ and radius r .

Figure: A 2-variable minimization problem with minimizer occurring on the boundary of the feasible region. Contour values are 0, 1, 2, 3, 4.

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- Graphically, we want to find the shortest distance of feasible points from the point $(4, 6)^T$.

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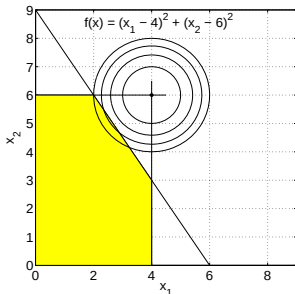


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- $(x_1 - 4)^2 + (x_2 - 6)^2 = r^2$ ($r > 0$) describes a circle with center $(4, 6)^T$ and radius r .
- Graphically, we want to find the shortest distance of feasible points from the point $(4, 6)^T$.
- Candidate: $3x_1 + 2x_2 = 18$.

Example 2.1 (cont'd)

$$\begin{array}{lll} \text{minimize} & f(\mathbf{x}) = (x_1 - 4)^2 + (x_2 - 6)^2 \\ \text{subject to} & x_1 & \leq 4 \\ & x_2 & \leq 6 \\ & 3x_1 + 2x_2 & \leq 18 \\ & x_1, x_2 & \geq 0. \end{array}$$

- We can reduce the 2-variable problem into a single variable optimization problem.
- The candidate boundary is the line $3x_1 + 2x_2 = 18$.
- Substituting $x_2 = 9 - \frac{3}{2}x_1$ into $f(\mathbf{x})$, we obtain a function in x_1

$$g(x_1) := f\left(x_1, 9 - \frac{3}{2}x_1\right) = (x_1 - 4)^2 + \left(3 - \frac{3}{2}x_1\right)^2, \quad 2 \leq x_1 \leq 4.$$

Example 2.1 (cont'd)

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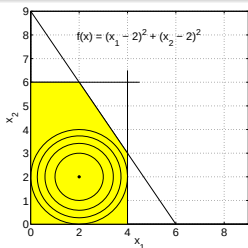
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- Use 1-variable calculus to determine a global minimizer of $g(x_1)$.
- Solving $g'(x_1) = \frac{13}{2}x_1 - 17 = 0$ gives $x_1^* = \frac{34}{13}$ and $x_2^* = \frac{66}{13}$.
- The other edges and vertices give larger values. Hence $\mathbf{x}^* := [\frac{34}{13}; \frac{66}{13}]$ is the global minimizer, with $f(\mathbf{x}^*) = \frac{468}{169}$.

Example 2.2 (Same as Example 2.1 but with a different objective function.)

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) := (x_1 - 2)^2 + (x_2 - 2)^2 \\ \text{subject to} & x_1 \leq 4 \\ & x_2 \leq 6 \\ & 3x_1 + 2x_2 \leq 18 \\ & x_1, x_2 \geq 0.\end{array}$$



• $\mathbf{x}^* := [2; 2]$ is global since $f \geq 0 = f(\mathbf{x}^*)$; it lies in the interior.

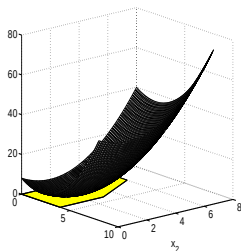


Figure: A 2-variable minimization problem with minimizer occurring in the interior of the feasible region. Contour values are 0, 1, 2, 3, 4.

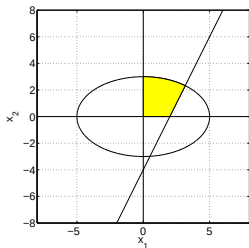
Figure: 3D plot.

Example 2.3 (A 2-variable NLP with a nonlinear objective function and a nonlinear constraint.)

$$\begin{array}{ll} \text{minimize} & f(x_1, x_2) := x_1^2 - x_2 \\ \text{subject to} & 2x_1 - x_2 \leq 4 \\ & 9x_1^2 + 25x_2^2 \leq 225 \\ & x_1, x_2 \geq 0. \end{array}$$

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The feasible region is bounded but not polygonal. For fixed x_1 , choose the largest feasible x_2 . Since $x_2 \leq 3$,

$$f(\mathbf{x}) \geq -3 = f(0, 3),$$

so $(0, 3)$ is a global minimizer.

Figure: A 2-variable minimization problem with a non-polygonal feasible region.

How good is the graphical method?

Pros:

- Good for intuition
- Provide rough solution

Cons:

- Directly practical mainly for problems with one or two variables
- Requires the computation of the function values on a dense grid of points is a costly task in general
- May need to rely on graph plotting toolbox

Discussions (cont'd)

Questions: for higher-dimensional problems

- How do we know whether the minimizer would occur on the boundary or in the interior of the feasible region?
- In the former case, on which part of the boundary?
- How do we know a given point $\mathbf{x}^*$ is a minimizer if we cannot visualize the graph of $f(\mathbf{x})$?

We will discuss these questions in topics 2, 4, 5.

Discussions (cont'd)

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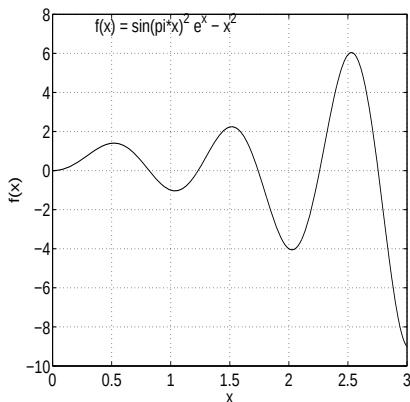
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It turns out that, **in general cases**, we may have either necessity or sufficiency, both never touch both.

Local vs global minimizers

Wait... what kind of minimizers do you mean?



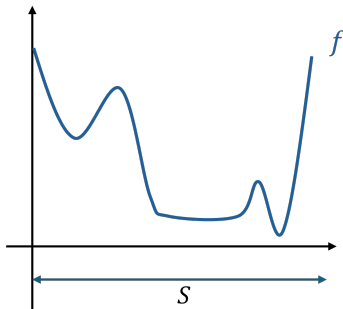
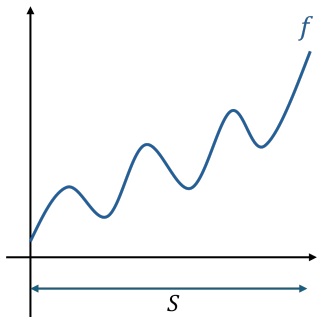
- What is the minimizer of f for the entire region $[0, 3]$?
- What is the minimizer of f for a small region $[1.8, 2.3]$?

Local minimizer and global minimizer

Definition 2.4 (Local minimizer and global minimizer)

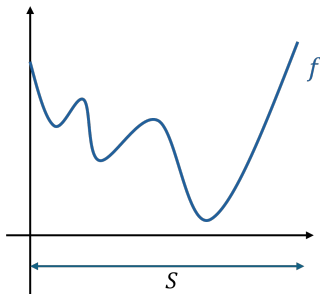
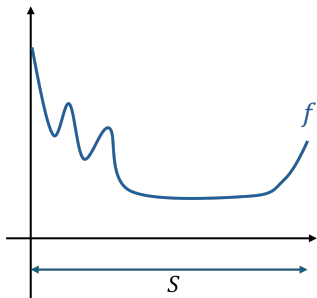
For $S \subseteq \mathbf{R}^n$, $f : S \rightarrow \mathbf{R}$, and $\mathbf{x}^* \in S$:

- The point $\mathbf{x}^*$ is a **local minimizer** if there exists $\epsilon > 0$ such that $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in S \cap B(\mathbf{x}^*, \epsilon)$.
- It is a **strict local minimizer** if the inequality is strict for every other feasible point in that neighborhood.



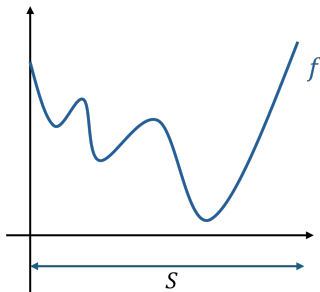
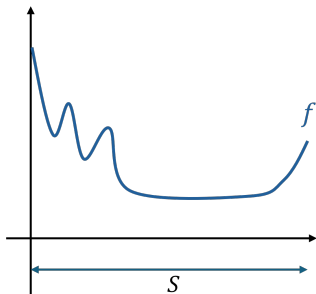
Definition 2.4 (Cont'd)

- A point $\mathbf{x}^* \in S$ is a **global minimizer** if $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for every $\mathbf{x} \in S$.
- It is a **strict global minimizer** if $f(\mathbf{x}^*) < f(\mathbf{x})$ for every $\mathbf{x} \in S \setminus \{\mathbf{x}^*\}$.



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Remarks.

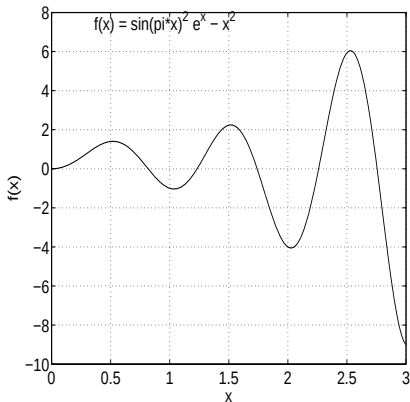
- Local and global maximizers are defined by reversing the inequalities.
- By definitions, a global minimizer is a local minimizer. However, the converse is not true in general.

Example 2.5

Consider the following 1-dimensional problem:

$$\min f(x) := \sin(\pi x)^2 \exp(x) - x^2$$

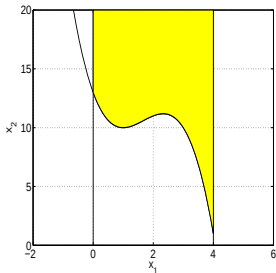
$$\text{s. t.} \quad 0 \leq x \leq 3$$



Example 2.6

Consider the following NLP in $\mathbf{R}^2$.

$$\begin{aligned} \min \quad & f(\mathbf{x}) := x_2 \\ \text{s. t.} \quad & 10 - (x_1 - 3)(x_1 - 1)^2 - x_2 \leq 0 \\ & 0 \leq x_1 \leq 4 \end{aligned}$$

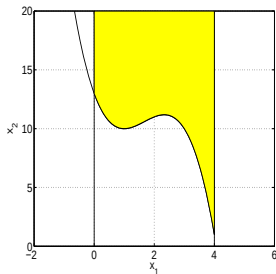


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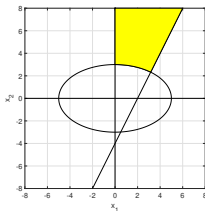
- 1 $\mathbf{x}^* := (4, 1)^T$ is a global minimizer.
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In general cases, we may target only local optimality.

Existence of global optimizers

Example 2.7 (Example 2.3 with a different feasible set)

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For every $t \geq 3$, the point $(0, t)$ is feasible and $f(0, t) = -t \rightarrow -\infty$; hence no global minimizer exists.

Case I: continuous function on a closed and bounded set

Definition 2.8 (Bounded)

A set $S \subseteq \mathbf{R}^n$ is **bounded** if there exists $M > 0$ such that $\|\mathbf{x}\| \leq M$ for every $\mathbf{x} \in S$.

Note: A set S is bounded if and only if there is a positive number $\hat{M}$ such that $|x_i| \leq \hat{M} \quad \forall i = 1, 2, \dots, n \quad \forall \mathbf{x} \in S$.

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Example 2.9

- (a) For $a < b$, the intervals (a, b) , $[a, b)$, $[a, b]$ and $(a, b]$ are bounded in $\mathbf{R}$.
- (b) The set $\{\mathbf{x} = (x_1, x_2, \dots, x_n)^T \in \mathbf{R}^n \mid a_i \leq x_i \leq b_i, i = 1, 2, \dots, n\}$ is bounded.
- (c) Every open or closed ball in $\mathbf{R}^n$ is bounded.
- (d) The set $C := \{(x, x^2) : x \in \mathbf{R}\}$ is not bounded. (Why?)

Case I: continuous function on a compact set

A closed set may not be a bounded set, eg: $S := \{x \in \mathbf{R} \mid x \geq 0\}$

A bounded set may not be a closed set, eg: $S := \{x \in \mathbf{R} \mid 0 < x < 1\}$.

We define a set with nice property:

Definition 2.10 (Compact)

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Definition 2.10 (Compact)

A set $S \subseteq \mathbf{R}^n$ is **compact** if it is closed and bounded.

It turns out we have a guarantee on the existence of global optimizers!

Theorem 2.11 (Weierstrass Theorem)

Let $S \subseteq \mathbf{R}^n$ be nonempty and compact, and let $f : S \rightarrow \mathbf{R}$ be continuous. Then f attains both a global minimum and a global maximum on S .

Example 2.12

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) := x_1^2 + 2x_2^2 \\ \text{s. t.} & x_1^2 + x_2^2 = 3.\end{array}$$

The feasible circle is compact, so Weierstrass guarantees both extrema. On it, $f = 3 + x_2^2$, so the minimum value is 3 at $(\pm\sqrt{3}, 0)$ and the maximum value is 6 at $(0, \pm\sqrt{3})$.

Example 2.13

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}) := x_1^2 + 2x_2^2 \\ \text{s. t.} & x_1 + x_2 = 1.\end{array}$$

The function f is continuous, but the feasible line is closed and unbounded. Thus, Weierstrass Theorem cannot be used directly to deduce whether there is a global minimum.

But the problem does have a global minimum: plugging $x_2 = 1 - x_1$ into f ,

$$f(\mathbf{x}) = x_1^2 + 2(1 - x_1)^2,$$

which has the unique minimizer $(x_1, x_2) = (\frac{2}{3}, \frac{1}{3})$.

Case II: coercive continuous function

What if S is unbounded?

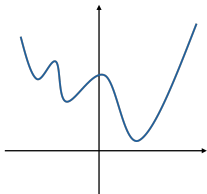
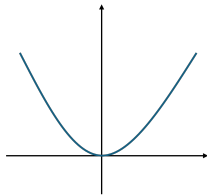
Definition 2.14 (Coercive function)

A function $f : \mathbf{R}^n \rightarrow \mathbf{R}$ is **coercive** if

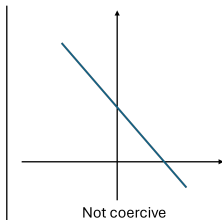
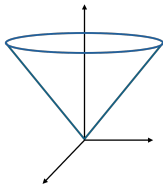
$$\|\mathbf{x}\| \rightarrow \infty \implies f(\mathbf{x}) \rightarrow +\infty.$$

Interpretation:

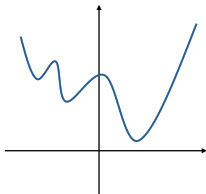
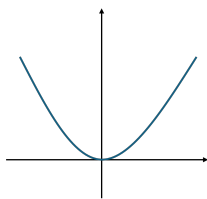
- $\|\mathbf{x}\| \rightarrow \infty$ means that $\mathbf{x}$ moves away from the origin
- the function value $f(\mathbf{x})$ of a coercive function increases without bound as $\mathbf{x}$ moves away from the origin, and in all possible directions



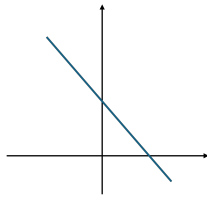
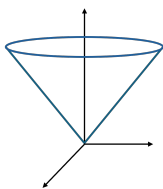
Coercive functions



Not coercive



Coercive functions



Not coercive

Example 2.15 (Single-variable coercive function)

Show that $f(x) := x^4 + 3x^2 + 3$ is coercive.

Solution. Since $\lim_{|x| \rightarrow \infty} f(x) = +\infty$, the function is coercive.

Example 2.16 (Multivariable functions)

- (i) $f_1 : \mathbf{R}^2 \rightarrow \mathbf{R}$, $f_1(\mathbf{x}) := x_1^2 + x_2^2 = \|\mathbf{x}\|^2$ is coercive.
- (ii) $f_2 : \mathbf{R}^2 \rightarrow \mathbf{R}$, $f_2(\mathbf{x}) := x_1^2$ is not coercive: $f_2(0, t) = 0$ while $\|(0, t)\| \rightarrow \infty$ as $t \rightarrow \infty$.

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- (iii) $f_3 : \mathbf{R}^2 \rightarrow \mathbf{R}$, $f_3(\mathbf{x}) := x_1^4 + x_2^4 - 3x_1^3 + x_2$ is coercive.

Proof. Let $r := \max\{|x_1|, |x_2|\}$. Since $\|\mathbf{x}\| \rightarrow \infty$ iff $r \rightarrow \infty$,

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Proof. Let $r := \max\{|x_1|, |x_2|\}$. Since $\|\mathbf{x}\| \rightarrow \infty$ iff $r \rightarrow \infty$,

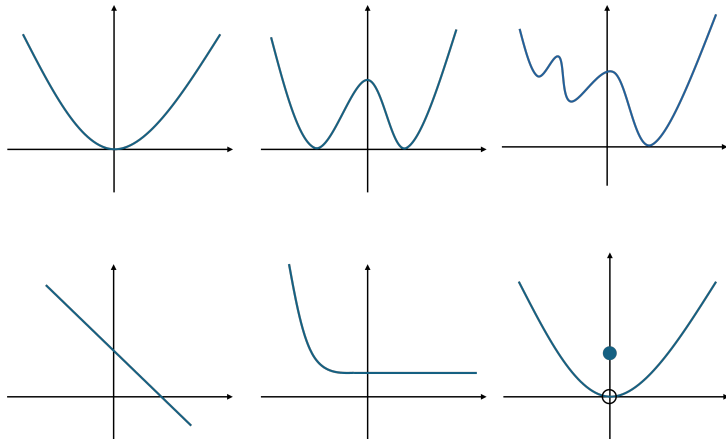
$$f_3(\mathbf{x}) \geq r^4 - 3r^3 - r \geq \frac{3}{5}r^4 \quad (r \geq 10),$$

which tends to $+\infty$.

Case II: coercive continuous function (cont'd)

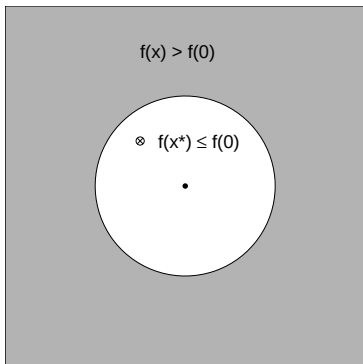
Theorem 2.17

Let $S \subseteq \mathbf{R}^n$ be nonempty and closed, and let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ be continuous and coercive. Then f attains a global minimum on S .



Idea for proving Theorem 2.17

Coercivity restricts the search to the intersection of S with a sufficiently large closed ball; Weierstrass applies there.



Take-home messages

- The graphical method for 1 or 2 variables, not for higher dimensions.
- Topologically defined local and global minimizers, our ultimate goals.
- Two sufficient conditions for existence of global optimizers.
- (Gossip: mathematicians have been struggling to close gaps everywhere; “no-free-lunch theorem”; be happy to deny yourselves :)

Analytic geometry

- René Descartes (1596-1650).
- Pierre de Fermat (1601-1665).

Extreme-value theorem

- Bernard Bolzano (1781-1848).
- Karl Weierstrass (1815-1897).

The list is definitely not comprehensive...